Course Code	Name of the Course					
216U03C602	RF Filters and Antennas					
Teaching Scheme	TH	P		TUT		Total
	03	--		--		03
Credits Assigned	03	--		--		03
Evaluation Scheme	Marks					
	LAB/TUT CA	CA (TH)		ESE		Total
		IA	ISE			
	--	20	30	50		100

Course pre-requisites:

- Electromagnetic Field theory
- Electrical Networks

Course Objectives:

The objective of this course is to enable the students to understand and analyze radio frequency behavior of transmission line and passive components like resistor, inductor and capacitor, design and analysis of filters by insertion loss method. The course covers the analysis of wire antennas, antenna arrays and special type of antennas like microstrip, helical, reflector needed for the communication system

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

CO1. Evaluate transmission lines parameters analytically and using smith charts.

CO2. Evaluate the high frequency behaviour of passive components and design different types of filters.

CO3. Understand the radiation characteristics of different wire antennas.

CO4. Analyse the radiation pattern of array.

CO5. Describe special type of antennas and their applications.

Detailed Syllabus

Module No.	Unit No.	Contents	Hrs	CO
1	**Introduction**		12	CO1
	1.1	Introduction to RF, RF spectrum and applications, two wire transmission line and it's equivalent circuit representation and transmission line parameters, types of transmission lines: Coaxial transmission line, parallel plate and microstrip transmission line.		
	1.2	Terminated lossless transmission lines, special termination conditions, source and loaded transmission line.		
	1.3	Introduction to smith chart, admittance transformation, transformation from reflection coefficient to load impedance using smith chart.		
	1.4	Impedance transformation using smith chart. #Self-Learning:RF hazards		
2	**RF Passive Components and Filter Design**		08	CO2
	2.1	RF behavior of passive components: Resistors, capacitors & inductors, introduction to chip components.		
	2.2	Filter parameters and configuration, Butterworth and Chebyshev type filter realization of low pass, high pass, band pass and band reject using insertion loss method for equal source and load resistance		
	2.3	Filter transformation, Impedance and frequency scaling, Microstrip filters design, Richard's transformation and Kuroda's identity. (Low pass filter design) Self Learning: S parameters		
3	**Fundamental Parameters of Antennas**		09	CO3
	3.1	Introduction to radiation mechanism, current distribution on thin wire antenna.		
	3.2	Antenna parameters: Radiation pattern, solid angle, radiation power density, radiation intensity, beamwidth, directivity, gain, antenna efficiency, beam area, beam efficiency, bandwidth , aperture concept, polarization, input impedance, antenna temperature, near and far field, Maximum directivity and maximum effective area, FRIIS transmission equation		
	3.3	Linear wire antennas: Potential functions, infinitesimal dipole, finite length dipole and half wave length dipole. (No derivations for Electric and Magnetic Field of wire antennas) #Self-Learning: Radiation integrals and auxiliary potential functions		

4	**Arrays**		**08**	**CO4**
	4.1	Introduction to antenna array, types and controlling parameters of array		
	4.2	Array of two isotropic point sources, non-isotropic sources and linear arrays of n elements: broadside and endfire array.		
	4.3	Principle of pattern multiplication and plot radiation pattern of an array.		
5	**Special Types of Antennas**		**08**	**CO5**
	5.1	Application of linear wire antennas: Monopole antenna, Yagi Uda antennas design and applications		
	5.2	Helical antennas: Modes of operation and design of axial mode helix.		
	5.3	Reflector antennas: Plane reflector, parabolic reflector and its feed mechanism.		
	5.4	Microstrip antennas: Rectangular microstrip antenna (RMSA) structure, transmission line model, advantages, disadvantages, applications and feed mechanism, design of RMSA.		
		Total	**45**	

Reference Books*

Sr No	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1	Reinhold Ludwig, Pavel Bretchko	*RF Circuit Design*	Pearson Education Asia, United States	2nd Edition, 2013
2	C. A. Balanis	*Antenna Theory Analysis and Design*	John Wiley & Sons Inc, India	3 rd Edition, 2005
3	John D. Kraus Ronald J. Marhefka, Ahmad S. Khan	*Antenna and wave Propagation*	Tata McGraw Hill publication, India	5 th Edition, 2017